DATABASES

Database Replication Visual Guide

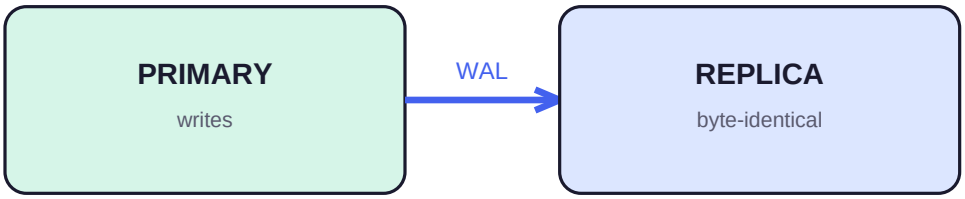
Streaming, lag, failover --
keeping copies in sync and surviving the primary's death.

ANIMATED WALKTHROUGH
Step through pages to play the animation

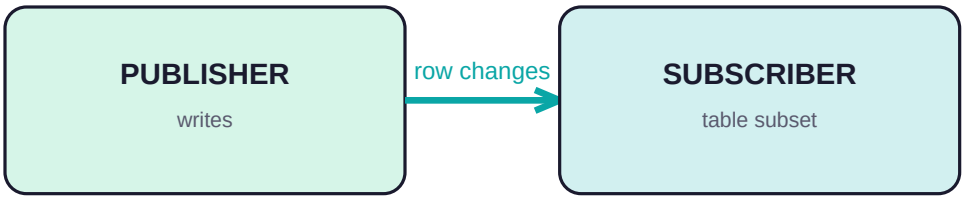
STEP 1/6

Streaming vs Logical Replication

Two ways to ship changes to a replica.



Physical (streaming): ships WAL bytes -- replica is a block-level clone.

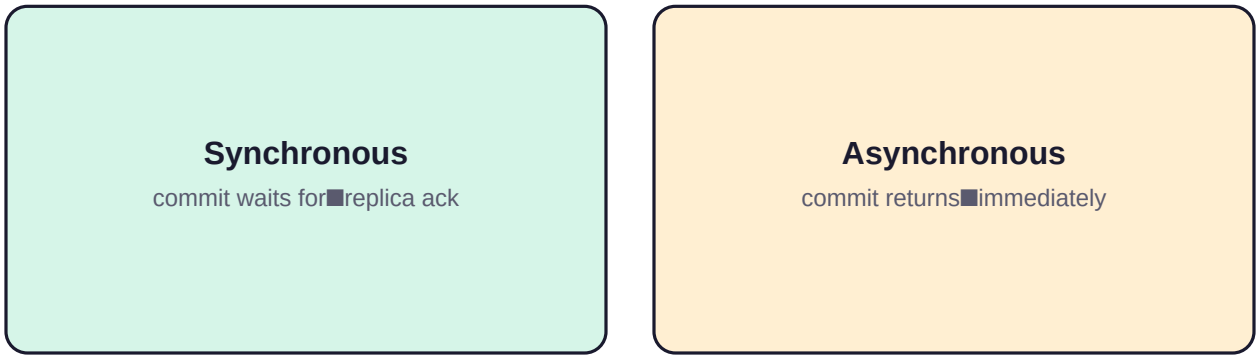


Logical: decodes row changes -- selective tables, cross-version, transforms possible.

STEP 2/6

Synchronous vs Asynchronous

Durability and latency are the same dial.



- Sync: zero data loss on failover -- but every write pays replica round-trip latency.
- Async: fast writes -- but a crash can lose the last seconds of transactions.
- Middle ground: sync to ONE nearby replica, async to the rest (quorum-style).

STEP 3/6

Lag Monitoring

A replica is only useful if it is current.

Replay lag (time)

how far behind in seconds -- alert at your RPO threshold

WAL bytes behind

volume of un-applied changes -- spikes signal trouble

Connection state

streaming / catchup / down -- dead replicas page loudly

Causes of lag: long-running queries on the replica, under-provisioned I/O, network partitions.

STEP 4/6

Failover & Promotion

The primary dies. Now what -- in order.

1. Confirm

primary truly dead, not partitioned -- avoid split-brain

2. Promote

replica becomes primary (pg_ctl promote / managed failover)

3. Repoint

apps connect to the new primary -- via proxy or DNS

4. Rebuild

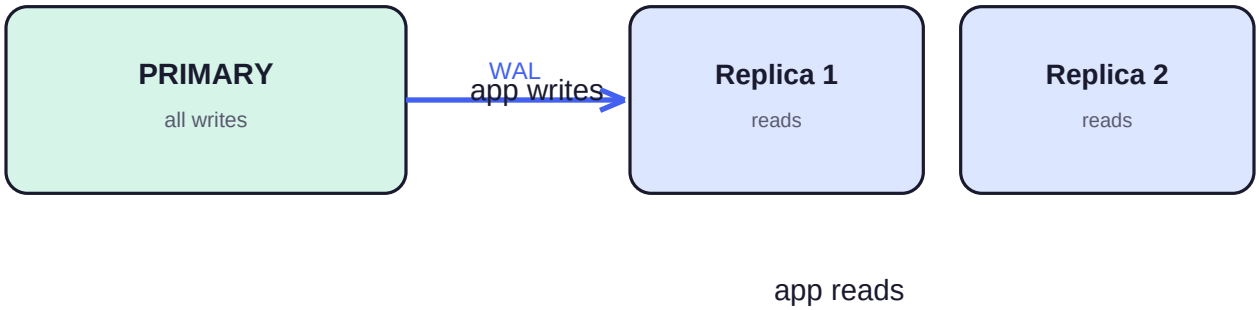
old primary rejoins as a replica -- or a fresh one is built

Practice this quarterly. Failover you have never rehearsed is failover that fails.

STEP 5/6

Read Scaling with Replicas

Offload reads -- carefully.



- Route analytics/reporting to replicas; keep transactional reads on the primary.
- Replica lag means read-your-write breaks: a user may not see their own update.
- Connection proxies (PgBouncer, ProxySQL) split read/write traffic transparently.

STEP ✓I✓

Key Takeaways

Database Replication -- Remember

- Physical replication clones WAL bytes; logical replication ships decoded row changes selectively.
- Synchronous = zero loss but slower writes; asynchronous = fast but can lose recent commits.
- Monitor replay lag in seconds AND bytes behind -- alert at your RPO threshold.
- Failover order: confirm death, promote, repoint apps, rebuild -- and rehearse it quarterly.
- Read replicas scale reads, but lag breaks read-your-write -- route wisely.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev